

Parameter-Efficient Fine-Tuning of PaliGemma for Remote Sensing Image Captioning

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Abstract—We apply QLoRA to PaliGemma-3B on the RISC remote-sensing captioning dataset, reaching CIDEr 2.21 from a zero-shot baseline of 0.25. Two ablations extend the recipe: a caption-selection strategy comparison, and a cumulative PEFT method comparison across vanilla LoRA, rsLoRA, LoRA+, and DoRA. A short exploration swaps the base to PaliGemma 2. The ablation suggests recent PEFT improvements do not stack monotonically on this domain; rsLoRA alone is the strongest single intervention at 500 steps.

I. INTRODUCTION

Vision-language models perform well at captioning but adapting them to specialized domains is expensive. QLoRA [2] addresses this with 4-bit base-weight quantization plus low-rank adapters [1]. We apply it to PaliGemma-3B [3], a SigLIP+Gemma VLM, on the RISC dataset of 44,521 satellite images each with five captions. Contributions: (i) a fine-tuned model with CIDEr 2.21 ($8.7\times$ baseline), (ii) caption-selection ablation, (iii) cumulative PEFT method comparison, (iv) a brief PaliGemma 2 swap.

II. APPROACH

The dataset is split image-level (seed 42) into 37,843 / 2,226 / 4,452 train/val/test. One of the five captions per image is sampled per training step (*random* strategy). We use `google/paligemma-3b-pt-224` with 4-bit NF4 double quantization and bf16 compute. LoRA adapters (rank 16, $\alpha=32$, dropout 0.05) attach to the seven attention and MLP projection modules, giving 22.6M trainable parameters (0.77%). The main run adds rsLoRA scaling [5] and LoRA+ [6] ($16\times$ LR on B matrices). AdamW at 10^{-4} , batch 4 with grad accumulation 4, gradient checkpointing, 3% warmup, 1 epoch (2,366 steps). Ablations share the recipe and cap at 500 steps. Metrics use pycocoEvalcap (multi-reference, all five captions).

III. RESULTS

Main result (Table I). CIDEr improves from 0.25 to 2.21; BLEU-4 from 0.04 to 0.58. Predictions shift from generic short outputs to longer, domain-specific captions (Fig. 1).

Caption strategy. At 500 steps, random sampling reaches CIDEr 1.83 (BLEU-4 0.53); picking only the first or longest caption falls to 1.42 (BLEU-4 0.47 and 0.53). Concatenating

Code: <https://github.com/sceran/paligemma3b-QLoRA-RISC-captioning>.
W&B: <https://wandb.ai/sceran/paligemma3b-QLoRA-RISC>, <https://wandb.ai/sceran/huggingface>.

TABLE I
BASELINE VS. QLoRA FINE-TUNED PALIGEMMA (500 TEST IMAGES).

	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr
Baseline	0.329	0.040	0.095	0.269	0.253
QLoRA tuned	0.853	0.585	0.409	0.750	2.207

TABLE II
CUMULATIVE PEFT METHOD ABLATION (500 STEPS EACH).

Configuration	BLEU-4	CIDEr
A. Vanilla LoRA	0.528	1.932
B. rsLoRA only	0.513	2.090
C. LoRA+ only	0.514	1.903
D. rsLoRA + LoRA+ stack	0.532	1.828
E. + DoRA	0.530	1.979
G. stack replicate	0.543	1.989

all five into one target collapses CIDEr to 0.00 because the model learns to produce over-long outputs the metric heavily penalizes.

Cumulative PEFT ablation (Table II). rsLoRA alone scores highest; rsLoRA+LoRA+ stack trails it; adding DoRA [7] recovers some ground. Two replicates of the stack (D and G) differ by 0.16 CIDEr, an empirical inter-run noise floor. Under this lens only rsLoRA marginally exceeds the noise band.

PaliGemma 2. The rsLoRA-only configuration on PaliGemma 2 3B [4] yields CIDEr 2.095 vs. PG1’s 2.090: a tie within the noise band. PG2 zero-shot is actually worse than PG1 (CIDEr 0.172 vs. 0.253), but fine-tuning closes the gap.

IV. DISCUSSION AND CONCLUSION

Modern PEFT improvements do not stack monotonically on RISC at this budget; with a 0.16 CIDEr noise floor from our replicates, vanilla LoRA stays competitive. The base-model swap alone does not buy quality: PG2 ties PG1 at a fixed budget, and the 1-epoch PG1 main run remains ahead of the 500-step PG2 run, so training duration outweighs v1→v2. QLoRA delivered an $8.7\times$ CIDEr gain with under 1% of the model trainable. Future work: full-epoch PG2, the instruction-tuned PG2 variant, and activation-driven initializations (EVA, CorDA) once their VLM paths stabilize.



Reference: A dense residential area has a grove of trees and lots of houses .
Baseline: aerial view of a house
Fine-tuned: The dense residential is next to the road and trees .



Reference: The industrial area has lots of big blue workshops and small white workshops .
Baseline: satellite image of the city
Fine-tuned: There are many houses of different shapes and sizes in the industrial zone .



Reference: many green trees are in a piece of forest with a pond .
Baseline: the view of the forest
Fine-tuned: many green trees are in a forest .



Reference: A dense residential area has lots of houses arranged neatly while some roads go through the residential area .
Baseline: satellite image of the city
Fine-tuned: The dense residential is next to the road .



Reference: a football field which is close to the busy street situates in a downtown area .
Baseline: satellite image of the city
Fine-tuned: a playground is surrounded by many buildings and a road .

Fig. 1. Five test images with reference, zero-shot baseline, and fine-tuned predictions. The baseline frequently falls back to generic phrasing (“satellite image of the city”, “aerial view of a house”), while the fine-tuned model produces RISC-style descriptions using domain vocabulary like “residential”, “industrial”, “forest”, and “playground”.

REFERENCES

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